

SP 7:2026 Impact Assessment

What the withdrawal of NBC 2016 means for Aptimizer's compliance engine, module by module.

The position

NBC 2016 was withdrawn on 30 April 2026 and replaced by SP 7:2026, the National Building Construction Standards. Aptimizer's compliance engine, its clause registry and its development controls all reference NBC 2016.

Two things follow, and they pull in opposite directions. The document Aptimizer implements no longer formally exists. But SP 7:2026 is reported to be voluntary until a state adopts it into its building bye-laws, and most states have not yet issued adoption notifications, so NBC 2016 remains what local approvals actually reference.

That is not a contradiction to resolve. It is the real position of every Indian practice right now, and the honest engineering response is to make the platform aware of both rather than to pick one.

Evidence caveat. *The withdrawal and the headline changes below come from secondary sources. SP 7:2026 does appear on the BIS standards portal, which corroborates that it exists. The specific clause values have not been read from the standard itself. Nothing in this assessment should be quoted as fact in a submitted report until it is checked against the BIS text or the gazette notification.*

Findings

Area	What changes	Impact on Optimizer	Sev.
Fire: high-rise threshold	The high-rise classification trigger is reported to move from 15 m to 24 m.	Every rule keyed to 15 m fires one band too early. A 16 to 23 m building is currently told it needs stairwell pressurisation and 50 000 L of static fire storage that SP 7 would not require. The platform is conservative rather than unsafe, but it is wrong, and it inflates cost and tank sizing for exactly the mid-rise projects most common in Tier 2 cities.	High
Fire: remaining thresholds	Part 4 folds into the restructured six-part document. Whether these specific values changed is not established from secondary sources.	Unknown until the clause text is read. Travel distance and refuge spacing are the two most likely to have moved, because both were revisited in the performance-based rewrite.	High
Clause registry: code identity	Twelve parts collapse to six across two volumes, and the name changes from Code to Standards.	Every NBC clause reference in the registry points at a withdrawn document, and the part numbering no longer maps. This also breaks the RAG corpus plan: a corpus built on NBC 2016 text will retrieve clauses that no longer exist under those numbers.	High
Development controls	Part 3 provisions move into the restructured document, and the language shifts from prescriptive shall to performance-based should.	Lower practical impact than it looks. The module's own docstring already says these are generally applied provisions that are stable across Indian jurisdictions, and setbacks are set by state and municipal bye-laws rather than by NBC directly. Those bye-laws have not changed.	Low
Prescriptive to performance-based	SP 7:2026 is reported to replace shall with should, moving design responsibility to the engineer.	This is the deepest change and it is architectural, not numerical. A performance-based standard has no single required value to check against, so a binary pass or fail and a percentage compliance score both stop being meaningful for any provision that becomes performance-based.	Structural
New coverage: EV, solar, decentralised STP	Parts D and E are reported to formally address EV charging, rooftop solar and decentralised sewage treatment.	Opportunity rather than a defect. All three already exist in the platform, sourced from guidelines rather than from the building code. If SP 7 now carries them, they can be cited to a standard instead.	Opportunity

Finding Detail: Evidence and Action

Fire: high-rise threshold [High]

In the code now:

`iscodes.FIRE['pressurisation_above_m'] = 15.0; FIRE_STATIC_STORAGE_L and FIRE_RESERVE_IN_SUMP_L both trigger above 15 m; engineering.m5_water uses tall = height > 15`

Action: Change the three 15 m constants to 24 m behind a code-version switch. Do not edit them in place: NBC 2016 is still what most state bye-laws reference.

Fire: remaining thresholds [High]

In the code now:

`max_travel_m 30.0, two_stair_height_m 24.0, refuge_above_m 24.0, refuge_every_floors 7, fire_lift_above_m 30.0`

Action: Read SP 7:2026 directly for each of these five values before claiming the module is current. Do not assume they carried over.

Clause registry: code identity [High]

In the code now:

`iscodes.CODE_LIBRARY` carries 19 entries citing NBC 2016 Part 3, Part 4, Part 8 and Part 9; 32 NBC 2016 string references across the file

Action: Add a `code_version` field to each registry entry rather than rewriting it. The app needs to cite whichever document the project's jurisdiction has actually adopted.

Development controls [Low]

In the code now:

`siteplan/devcontrols.py` carries 13 NBC 2016 Part 3 references: the height-driven open space table, front setback by plot area, height versus road width

Action: Leave as is. Note in the report that setbacks derive from local bye-laws, which continue to reference the older provisions.

Prescriptive to performance-based [Structural]

In the code now:

`engineering.check()` returns a binary pass or fail against a fixed required value; compliance scores are computed as `passed / total x 100`

Action: No code change yet. Recognise it as a limitation, and say so: the compliance engine is built on a prescriptive model, and a performance-based standard would need a different engine, not different constants.

New coverage: EV, solar, decentralised STP [Opportunity]

In the code now:

EV at `PARKING['ev_pct'] = 20.0` citing MoHUA and BEE 2019; solar potential in `gis.py`; STP selection in `m5_water`

Action: Once you have the text, re-source these three from SP 7 and keep the MoHUA and BEE citations as the fallback.

Live Defect Found During This Audit

Unrelated to SP 7:2026, but found while tracing every NBC reference and worth fixing before anyone reviews the code.

Clause library contradicts the engine on travel distance

Evidence

`iscodes.CODE_LIBRARY` entry `nbc4` states `key_value 'Travel 22.5 m; 2 stairs > 24 m ...'`, while `iscodes.FIRE['max_travel_m'] = 30.0` and `m8_fire` checks against `30.0`. A comment above `FIRE` records that `22.5` was the old wrong value and was corrected to `30.0`.

Why it matters. The constant was fixed but the library card was not. The IS/NBC Engineering module reads `CODE_LIBRARY`, so a user reading the clause card is told `22.5 m` while the compliance check beside it uses `30 m`. Two numbers for one rule on one screen is exactly the failure the single-source-of-truth design exists to prevent, and an examiner who spots it will reasonably ask what else drifted.

Fix. Change the `nbc4` `key_value` to read `30 m`. Better: derive the `key_value` string from the `FIRE` constants so it cannot drift again, the same way the `m8_fire` check labels already do.

Recommended approach

Do not rewrite the constants in place. Add a code-version dimension instead.

- Add a `code_version` field to the project, defaulting to `nbc2016`, with `sp7_2026` as the alternative.
- Key the constant tables on it, so `FIRE` becomes a lookup by version rather than a single dict. The single-source-of-truth design in `iscodes.py` already makes this a contained change.
- Report the version on every compliance output and in every generated report, so a reader always knows which document a result was checked against.
- Leave SP 7 values unset until you have read the standard. An unset value should refuse to check rather than fall back silently to the 2016 number, which is the same discipline the code corpus already applies when it has no clause text.

What to say in the viva. That the platform implements NBC 2016 because that is what state bye-laws currently reference; that SP 7:2026 has withdrawn it at national level; that you have assessed the migration and know which thresholds move; and that the compliance engine is prescriptive by design, which a performance-based standard will eventually require rethinking. Being the person who raises this is a considerably stronger position than being the person told about it.